

A
Mini Project Report
on
“SuperMarket Billing System”

Presented by

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Submitted to



MGM's College of Engineering, Nanded

Under

Dr. Babasaheb Ambedkar Technological University, Lonere

Certificate

This is to certify that the mini project entitled

“Super Market Billing System”

Submitted By

Purva Mukund Gharapurkar

In satisfactory manner as a partial fulfillment of

SY[CSE-A] in Computer Science and Engineering

To

MGM’s College of Engineering, Nanded

Under

Dr. Babasaheb Ambedkar Technological University, Lonere

has been carried out under my guidance,

Ms. N. L. Pariyal / Ms. S. S. Wagre

Mini Project Guide

Title = “Super Market Billing System”

Abstract ==>

This project, Supermarket Billing System, is a Java-based console application that automates the billing process in supermarkets. It displays a list of products, allows the customer to enter product numbers and quantities, calculates the total bill, applies discounts based on the bill amount, and finally prints a detailed bill summary. The project demonstrates concepts like arrays, loops, conditional statements, and formatted output in Java. This system helps reduce human errors and ensures fast and accurate billing.

Introduction ==>

The Supermarket Billing System is designed to simplify billing operations in retail stores. It automates the calculation of product totals, applies discounts, and generates a final bill. The system works using basic Java programming concepts and is easy to use for both customers and shop owners. This mini-project provides practical experience with array handling, user input, conditional operations, and structured program logic.

Objective ==>

- To develop an automated billing system.
- To reduce manual calculation errors.
- To apply discount rules based on total purchase value.
- To generate a clear and readable bill format.
- To understand Java basics through real-life application.

Algorithm ==>

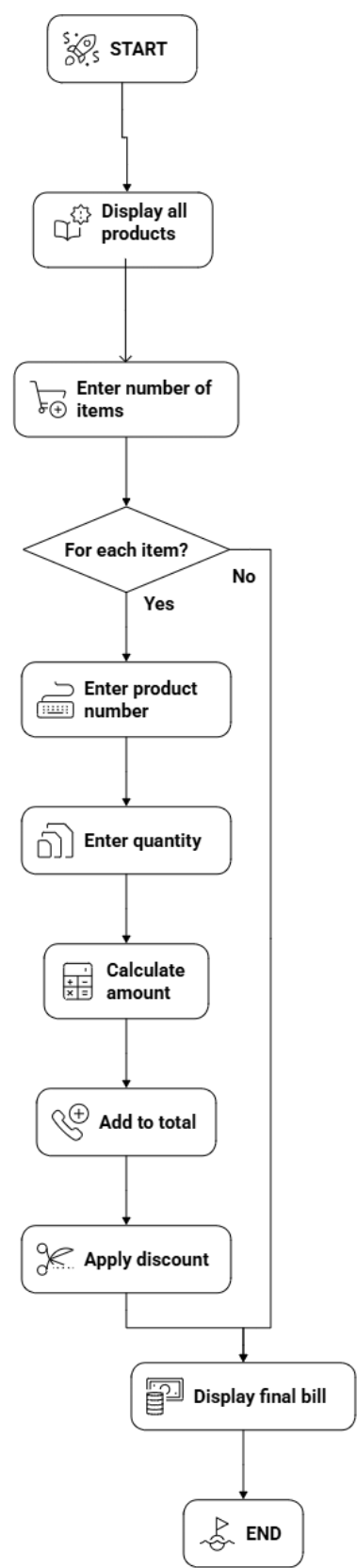
1. Start the program.
2. Store product names and prices in arrays.
3. Display all products with serial numbers.
4. Ask the user how many different items they want to buy.
5. For each item:
 - Read product number
 - Read quantity
 - Calculate item total = price × quantity
 - Add item total to overall total
6. Apply discount based on the total bill amount.

7. Display the final formatted bill.
8. End the program.

Working ==>

- The system displays a menu of 39 products.
- The user selects items by entering product numbers.
- Quantity is entered for each selected product.
- The system calculates the total price and applies discounts:
 - Above ₹500 → 10%
 - Above ₹200 → 5%
- Finally, a bill containing product name, quantity, price, item total, and grand total is generated.

Flowchart ==>



Program ==>

```
import java.util.Scanner;

public class SuperMarketBillSystem {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String[] products = { "Rice 1kg", "Wheat Flour 1kg", "Sugar 1kg", "Salt 1kg", "Tur Dal 1kg", "Moong Dal 1kg", "Chana Dal 1kg", "Oil 1L", "Ghee 500g", "Milk 1L", "Bread", "Butter 100g", "Tea Powder 250g", "Coffee 200g", "Biscuits", "Noodles Pack", "Poha 1kg", "Rava 1kg", "Cornflakes 100 gm", "Shampoo 180ml", "Soap 75 gm", "Toothpaste", "Toothbrush", "Detergent 1kg", "Dishwash Liquid 155 ml", "Handwash 250ml", "Sanitizer 100ml", "Tissue Pack", "Paneer 200g", "Curd 500g", "Chips Pack", "Cold Drink 1L", "Chocolate", "Ketchup 500g", "Pickle 200g", "Dry Fruits 250g", "Ice Cream Cup", "Batteries (2pc)", "Matchbox"};

        double[] prices =
{115,61.50,44.50,27,101,179,111,185,39,56,50,55,137,123,10,40,70,78,53,90,75,61,45,64.97,20,110,50,69,81,
24,10,70,20,105,37,120,25,66.8,8};

        int [] quantity = new int[products.length];

        System.out.println("Welcome to the Super Market!");

        System.out.println("\n Here is the list of products available:");

        System.out.println("===== SUPERMARKET MENU =====\n");

        for (int i=0; i<products.length; i++) {

            System.out.println((i + 1) + ". " + products[i] + " - Rs. " + prices[i]);

        }

        System.out.print("\nHow many different items do you want to purchase? ");

        int n = sc.nextInt();

        for (int i = 0; i < n; i++) {

            System.out.print("\nEnter product number (1-" + products.length + "): ");

            int p = sc.nextInt();

            int index = p - 1;

            System.out.print("Enter quantity of " + products[index] + ": ");

            quantity[index] += sc.nextInt();

        }

    }

}
```

```
}
```

```
// Generate Bill
```

```
System.out.println("\n===== BILL =====");
```

```
double total = 0;
```

```
System.out.printf("%-25s %-10s %-10s %-10s\n", "Product", "Quantity", "Price", "Total");
```

```
System.out.println("-----");
```

```
for (int i = 0; i < products.length; i++) {
```

```
    if (quantity[i] > 0)
```

```
{
```

```
    double itemTotal = prices[i] * quantity[i];
```

```
    total += itemTotal;
```

```
    System.out.printf("%-25s %-10d %-10.2f %-10.2f\n", products[i], quantity[i], prices[i], itemTotal);
```

```
}
```

```
}
```

```
System.out.println("-----");
```

```
System.out.printf("Grand Total: Rs. %.2f\n", total);
```

```
// Apply Discounts
```

```
double discount = 0;
```

```
if (total > 500)
```

```
{
```

```
    discount = total * 0.10; // 10% discount
```

```
    System.out.println("Discount Applied: 10%");
```

```
}
```

```
else if (total > 200)
```

```
{
```

```

        discount = total * 0.05; // 5% discount

        System.out.println("Discount Applied: 5%");

    }

    else

    {

        System.out.println("Shop more than Rs.200 or Rs.500 to get exciting discounts!");

        discount = 0;

    }

    double finalAmount = total - discount;

    System.out.printf("Discount Amount: Rs. %.2f\n", discount);

    System.out.printf("Final Amount to Pay: Rs. %.2f\n", finalAmount);

    System.out.println ("Thank you for shopping with us!");

    sc.close();

}

}

```

Output ==>

Welcome to the Super Market!

Here is the list of products available:

===== SUPERMARKET MENU =====

1. Rice 1kg - Rs. 115.0
2. Wheat Flour 1kg - Rs. 61.5
3. Sugar 1kg - Rs. 44.5
4. Salt 1kg - Rs. 27.0
5. Tur Dal 1kg - Rs. 101.0
6. Moong Dal 1kg - Rs. 179.0
7. Chana Dal 1kg - Rs. 111.0

8. Oil 1L - Rs. 185.0
9. Ghee 500g - Rs. 39.0
10. Milk 1L - Rs. 56.0
11. Bread - Rs. 50.0
12. Butter 100g - Rs. 55.0
13. Tea Powder 250g - Rs. 137.0
14. Coffee 200g - Rs. 123.0
15. Biscuits - Rs. 10.0
16. Noodles Pack - Rs. 40.0
17. Poha 1kg - Rs. 70.0
18. Rava 1kg - Rs. 78.0
19. Cornflakes 100 gm - Rs. 53.0
20. Shampoo 180ml - Rs. 90.0
21. Soap 75 gm - Rs. 75.0
22. Toothpaste - Rs. 61.0
23. Toothbrush - Rs. 45.0
24. Detergent 1kg - Rs. 64.97
25. Dishwash Liquid 155 ml - Rs. 20.0
26. Handwash 250ml - Rs. 110.0
27. Sanitizer 100ml - Rs. 50.0
28. Tissue Pack - Rs. 69.0
29. Paneer 200g - Rs. 81.0
30. Curd 500g - Rs. 24.0
31. Chips Pack - Rs. 10.0
32. Cold Drink 1L - Rs. 70.0
33. Chocolate - Rs. 20.0
34. Ketchup 500g - Rs. 105.0

35. Pickle 200g - Rs. 37.0

36. Dry Fruits 250g - Rs. 120.0

37. Ice Cream Cup - Rs. 25.0

38. Batteries (2pc) - Rs. 66.8

39. Matchbox - Rs. 8.0

How many different items do you want to purchase? 5

Enter product number (1-39): 35

Enter quantity of Pickle 200g: 1

Enter product number (1-39): 33

Enter quantity of Chocolate: 2

Enter product number (1-39): 39

Enter quantity of Matchbox: 3

Enter product number (1-39): 37

Enter quantity of Ice Cream Cup: 1

Enter product number (1-39): 31

Enter quantity of Chips Pack: 1

===== BILL =====

Product	Quantity	Price	Total
---------	----------	-------	-------

Chips Pack	1	10.00	10.00
------------	---	-------	-------

Chocolate	2	20.00	40.00
Pickle 200g	1	37.00	37.00
Ice Cream Cup	1	25.00	25.00
Matchbox	3	8.00	24.00

Grand Total: Rs. 136.00

Shop more than Rs.200 or Rs.500 to get exciting discounts!

Discount Amount: Rs. 0.00

Final Amount to Pay: Rs. 136.00

Thank you for shopping with us!

Conclusion ==>

The Supermarket Billing System is a simple yet effective application that demonstrates the power of Java programming in real-life scenarios. It automates billing, applies discounts efficiently, and reduces effort for shop owners. The project successfully meets all objectives and can be further expanded into a full supermarket management system.